Craig Breen-Hogg

XR Developer | Technical Artist | Unity / Unreal
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LinkedIn | ArtStation | Rookies | Github

Multi-disciplinary developer with 4+ years' experience in XR/VR, real-time pipelines, and technical art. Skilled in Unreal Engine, Unity, and Python, with a background in building interactive experiences, simulation content, and optimised 3D assets. Contributed to games including Ashborn (FireSlug Studios) and Samsara (Infinity27), with a focus on performance, scalability, and real-time integration.

Software Experience & Skills

Maya/Blender Copilot Studios & Claude GitHub/Perforce
Substance Painter Adobe Suite Zbrush
Unreal Engine/Unity Gaea Trello
Project Management C# Altoura
Blueprinting Fusion 360 Photogrammetry
Shaders Python HTML/CSS

Employment History:

Unity Developer/3D Artist, SCG.World - August 2022 to Present
SCG.World, a Microsoft Mixed Reality Partner, specialises in developing innovative solutions across multiple industries, focusing on HoloLens and simulator technologies.
Designed and implemented Azure Digital Twins (ADT) environments in Unity, simulating real-world IoT-enabled sites and data flows.
Developed promotional material in Unreal Engine, securing contracts with Sheffield Super Trams and The Crown Estate.
Developed real-time XR applications in Unreal Engine and Unity for clients across rail, healthcare, oil & gas, and defence sectors.
Lead R&D for photogrammetry workflows using tools like Recap Pro, Meshroom, and Pixyz.
Enhanced train simulator engines for Avanti West Coast, optimizing 3D assets for VR/HoloLens.
Worked with HoloLens, Meta Quest Pro, and Meta Quest 3 devices.
Developed an POC AR experience for The Crown Estate in less than a month.
Created custom training scenarios with Altoura all the way through to final implementation.
Supported end-to-end deployment, including device configuration, client setup, and technical troubleshooting for XR hardware.
Collaborated directly with clients to translate technical requirements into interactive solutions, iterating on performance and usability.
Applied 3D optimisation techniques (LODs, draw call reduction, texture streaming) to improve performance across XR platforms.
Contributed to development of a prostate biopsy simulation, focusing on interaction systems, real-time feedback, and visual fidelity.
Built and maintained web-based solutions (HTML, CSS, JavaScript) to support client delivery, internal tools, and product showcases.
AI Applied Engineer/Forward Deployed Engineer, DCH Cottages (Contract) - January 2026 to Present
Worked extensively with the Marketing Manager, Property Manager & Owner to build MVPs then ship them as usable products to enhance their business & processes.
Worked directly with stakeholders to elicit requirements, understand operational challenges, and translate business needs into technical solutions.
Designed and built AI-powered applications, including chatbots and website widgets, tailored to specific

business requirements.
Developed automation pipelines for review management and lead-generation workflows, streamlining business processes and reducing repetitive manual tasks.
Applied AI and software development to practical business problems, combining requirements gathering, solution design, implementation, and iterative improvement.
Developed web-based applications and tools that connect business requirements with functional, user-focused software.
Founder & Software Developer, Proper Useful - January 2026 - Present
Founded and developed a digital solutions business specialising in SaaS platforms, lead generation calculators, custom website development, interactive widgets and AI-powered chatbots. Building practical, data-driven tools that help businesses attract customers, generate qualified leads and improve online performance.
Designed and developed interactive calculators for the holiday let, green energy and wedding venue industries, helping owners and investors estimate revenue potential, occupancy, profitability and return on investment.
Built conversion-focused calculators and digital tools designed to capture enquiries, qualify potential customers and turn website visitors into actionable business leads.
Designed and built bespoke websites tailored to individual business requirements, with a focus on user experience, functionality, performance and conversion optimisation.
Developed embeddable widgets and custom web applications that provide interactive functionality, streamline user journeys and add value to existing websites. Such as a what's nearby widget for DCH Cottages utilising the Geoapify API.
Developed AI-powered chatbot solutions to automate customer interactions, answer enquiries and support lead capture and qualification.
Utilised JavaScript, HTML, CSS and Python to build responsive interfaces, calculation engines and dynamic web functionality.
Implemented dynamic pricing, occupancy modelling and financial forecasting logic to simulate real-world scenarios and deliver useful, data-driven insights.
Managed the full development lifecycle, from initial concept and requirements through design, development, testing, deployment, maintenance and ongoing improvements.
Created landing pages and optimised user journeys to improve engagement, encourage enquiries and support customer acquisition.
Designed and developed a bespoke review automation hub for a client, streamlining the collection, processing and publishing of customer reviews through API integrations, web scraping testing and automated parsing of copied data and Airbnb exports.
Core Technologies & Expertise: JavaScript, HTML, CSS, Python, SaaS Development, Full-Stack Web Development, Financial Modelling, Lead Generation, AI Chatbots, API Integrations, Website Development and Conversion Rate Optimisation.
Technical Artist, Infinity27 (Contract) - July 2024 to November 2024
Infinity27 is a games studio based in Gateshead that are currently working to develop their title Samsara which is currently out on early access.
Designed and developed a PCG Tool to help environment artists and level designers efficiently create levels and add intricate details.
Ported the tool from Unreal Engine 5.2 to 5.4, leveraging new PCG features introduced in the update.
Collaborated with the team's artists to enhance tool usability, adding parameters like scale, rotation, and offset controls for greater flexibility.
Contributed to materials, Niagara, lighting, optimization, and blueprint systems to support various aspects of the project.

Strengthened professional practices with project management tools, version control, and consistent organization and naming conventions.

VR Environment Artist, UXey Ltd. - January 2022 to February 2024

Designed and built complete VR environments in Unreal Engine, focusing on real-time performance, lighting, and optimisation

Implemented level design, asset integration, and blueprint-based interaction setups for immersive experiences

Collaborated directly with stakeholders to define requirements and guide project direction, ensuring delivery milestones were met

3D Art Team Lead, FireSlug Studios - August 2021 to May 2023

Managed a team of 10 3D artists and animators, coordinating delivery across sprint cycles using Agile workflows

Integrated and optimised assets within Unity, ensuring compatibility with real-time rendering and gameplay systems

Defined and enforced asset pipelines, including LOD generation, polygon budgets, and memory optimisation strategies

Provided technical guidance and mentoring on 3D workflows, improving team output quality and efficiency

Developed and maintained an Art Bible, establishing style, technical standards, and production consistency

Established environment metrics and modular design guidelines for scalable asset creation

Used Trello for sprint planning, task tracking, and cross-team coordination

Supported team scaling by recruiting, onboarding, and integrating 5 new artists into production workflows

Freelancer, CBH Studios - January 2021 to January 2022

Integrated assets into Unreal Engine, ensuring compatibility with real-time rendering and interaction systems

Initiated and prototyped a client VR application, contributing to interaction logic and scene setup

Managed end-to-end technical delivery, including client requirements, iteration cycles, and deployment readiness

Education

MA Indie Games Design, Falmouth University Merit - 2021 to 2023

BA (Hons) Games Design, Falmouth University (Cornwall College) 1st Class - 2020 to 2021

Other Experience

Infinity27 Tech Art Skills Bootcamp. Unity VR Development Pathway.

100 Days of Python Udemy Course. Unity Creative Core Pathway

Unity Junior Programmer Pathway. Enterprise Agent Challenge (Copilot Studio)

Personal Projects/Testing

AI Discord Productivity Bot - Python, AI Api, Obsidian

Dating App - React

Sim Racing Map Generator - HTML, CSS, Unreal Engine 5, PCG, C++, Blueprints, Shaders

Golf Launch Monitor (Software Testing Only) - Python, HTML

Tank VR Diorama (Collaberation) - Unity, C#

Green Energy Calculator (Proper Useful) - Python, HTML, CSS

Tattup Tattoo App - Unity, C#, React

Whatsapp Chat Bot Agent - Python

Surveyancing First Pass Application - HTML, CSS, Python

Golf Bag Builder Website - HTML, CSS, Webscraping, Python

NERF/Gaussian Splat Editing Tool - HTML, ThreeJS, 3D